

Nassau Candy Factory Reallocation & Shipping Optimization

An Operations Research Case Study & Data Science Implementation Guide

Target Audience: Operations Strategy & Data Science Students Subject: Supply Chain Predictive Analytics Status: Implementation Guide

1. Executive Summary

In modern logistics networks, fulfilling geographic demand efficiently depends heavily on assigning product groups to optimal manufacturing facilities. This case study details the Nassau Candy Factory Reallocation & Shipping Optimization Framework, a data-driven system built to analyze multi-facility order patterns, predict logistics delays, and evaluate tactical supply chain reassignments.

By transforming historical order parameters into predictive features, the framework helps supply chain executives shift from reactive scheduling to predictive logistics. It balances two competing objectives: reducing transit times and preserving profitability. Through a predictive champion model, a scalable scenario simulation engine, and an interactive dashboard interface, this project acts as an operational playbook for warehouse and production network optimization.

Δ Lead Time PRIMARY METRIC	Risk-Adjusted PROFIT SAFEGUARD	Multi-Model LR / RF / GBM	Streamlit UI DECISION SUPPORT
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2. Business Context & Problem Statement

Nassau Candy operates multiple manufacturing and distribution facilities nationwide, serving a diverse base of commercial accounts across various regions. Currently, product lines are mapped to specific factories based on legacy constraints and historical production capabilities. This static mapping creates notable supply chain friction:

- **Geographic Mismatch:** Products are frequently manufactured in distant facilities and cross-shipped across regions, increasing transit times and shipping costs.
- **Unpredictable Lead Times:** Seasonal variation, localized facility bottlenecks, and varying regional carrier performance lead to inconsistent shipping cycles.
- **Margin Erosion:** Expedited shipping choices (e.g., Next-Day vs. Standard Ground) hide structural flaws in facility assignments, turning production savings into shipping expenses.

The core challenge is to build an analytical framework that answers a crucial operational question: *If we reallocate the production of Product X from Factory A to Factory B, what will be the true impact on shipping lead times, margin security, and delivery stability?*

3. Data Schema & Feature Engineering

The foundational data comes from the historical order matrix `Nassau Candy Distributor.csv`. The operational pipeline extracts, cleans, and builds features from these records to support predictive modeling.

3.1 Primary Fields and Engineered Attributes

- **Temporal Columns:** `Order Date` and `Ship Date` are parsed to derive the core operational target variable:

$$\text{ext}\{\text{Lead Time Days}\} = \text{ext}\{\text{Ship Date}\} - \text{ext}\{\text{Order Date}\}$$

- **Categorical Variables:** `Product Name`, `Product ID`, and `Division` define production constraints. `Region`, `State/Province`, `City`, and `Postal Code` map out geographical demand. `Ship Mode` represents the transportation tier.
- **Financial Attributes:** `Sales`, `Units`, `Gross Profit`, and `Cost` are used to engineer a margin metric for profitability filtering:

$$\text{ext}\{\text{Profit Margin}\} = \text{rac}\{\text{ext}\{\text{Gross Profit}\}\{\text{ext}\{\text{Sales}\}\}$$

3.2 Data Preprocessing & Robustness

Real-world order logs often contain anomalies like clerical errors or data entry mistakes. To keep predictive models accurate and more stable during training, the data preparation script applies outlier clipping to key target and continuous financial attributes.

Rather than standard deletion, a robust winsorization clipping strategy is used. Elements falling outside the 1st and 99th percentiles for the feature set $C = \{\text{ext}\{\text{lead_time_days}\}, \text{ext}\{\text{Sales}\}, \text{ext}\{\text{Units}\}, \text{ext}\{\text{Gross Profit}\}, \text{ext}\{\text{Cost}\}\}$ are dynamically clamped to their respective lower and upper quantile thresholds:

$$x_{\{\text{ext}\{\text{clipped}\}\}} = \max\left(Q_{\{0.01\}}(x), \min\left(Q_{\{0.99\}}(x), x \text{ ight}\right) \text{ ight}\right)$$

Furthermore, rows with missing values across these critical dimensions are eliminated, and only structurally valid records where $\text{ext}\{\text{lead_time_days}\} \geq 0$ are preserved. This prevents negative time distortions from affecting gradient descents or split determinations in decision trees.

```
# Implementation reference from nassau_optimization/data.py
def _clip_outliers(df: pd.DataFrame, columns: Iterable[str]) -> pd.DataFrame:
    # Clip extreme values to the 1st and 99th percentiles to reduce noise
    clipped = df.copy()
    for column in columns:
        lower = clipped[column].quantile(0.01)
        upper = clipped[column].quantile(0.99)
        clipped[column] = clipped[column].clip(lower, upper)
    return clipped
```

4. Predictive Modeling Methodology

Instead of relying on historical averages, this framework uses regression algorithms to predict delivery lead times based on order profiles.

4.1 Time-Based Validation Split

Traditional random k-fold cross-validation can introduce future data leakage when working with logistics data. To prevent this, the framework splits data chronologically. Models are trained on early historical records and tested on the most recent orders, mirroring real-world deployment.

4.2 Model Evaluation Matrix

The system trains and evaluates three algorithms to ensure robust performance across various data types:

1. **Linear Regression:** Establishes a baseline for clear, predictable coefficients.
2. **Random Forest Regressor:** Captures nonlinear interactions between product categories, regions, and distribution centers.
3. **Gradient Boosting Machine (GBM):** Iteratively corrects residual errors, often delivering the highest predictive accuracy for subtle patterns.

The champion model is selected by evaluating three key performance metrics:

- **Root Mean Squared Error (RMSE):** Penalizes large errors to avoid costly fulfillment delays.
- **Mean Absolute Error (MAE):** Represents the average expected delay variance in days.
- **Coefficient of Determination (R²):** Indicates the percentage of lead time variability explained by the features.

5. Scenario Simulation Engine

The core logic of the optimization framework is contained within `nassau_optimization/simulation.py`. It performs systematic "what-if" evaluations for candidate product-to-factory reassignments.

Simulation Routine Execution Pattern

For a selected target product currently mapped to a baseline factory, the simulation engine swaps the Factory identifier attribute for all historical order instances with an alternative candidate facility, keeping the destination region and ship mode constant.

The engineered features are fed into the champion machine learning model to generate a new synthetic lead time forecast. The alternatives are then scored using three criteria:

- **Lead Time Reduction (%):** Calculated as the percentage change between original and simulated lead times:

$$\Delta LT = \frac{LT_{\text{baseline}} - LT_{\text{simulated}}}{LT_{\text{baseline}}} \times 100$$

- **Profit Impact Proxy:** Tracks gross profit stability to ensure that routing shifts do not negatively affect operational margins.
- **Risk Score:** Derived from historical variance. If an alternate path has high variability in historical data, its confidence score decreases:

$$\text{Confidence Score} = \frac{1}{\sigma_{\text{historical}}^2}$$

6. Interactive Decision Support Dashboard (Streamlit)

To make these complex backend models accessible to business users, the framework includes an interactive Streamlit application (`app.py`). This dashboard translates data science metrics into actionable operational insights.

6.1 Core Dashboard Components

- **Control Sidebar:** Allows users to filter by specific product families, destination markets, and carrier ship modes. A custom optimization slider lets users adjust priorities between fulfillment speed and margin protection.
- **KPI Cards Block:** Displays key operational metrics, including average baseline lead times, expected time reductions, and overall scenario confidence.
- **Ranked Recommendations Panel:** Automatically generates an ordered list of factory assignment opportunities, flagging high-risk options based on model uncertainty.
- **Data Export Functionality:** Generates down-stream execution schedules, allowing warehouse managers to export optimized allocation strategies directly to CSV.

7. Suggested Student Deliverables & Research Directions

Students working through this case study can format their findings into three clear deliverables:

Deliverable Element	Core Objective	Technical Target
1. Research Write-Up	Document EDA, feature selection, and algorithm performance.	Compare baseline R ² against ensemble methods.
2. Streamlit Dashboard	Deploy a functional app with filters and interactive data tables.	Maintain separation between UI logic and simulation modules.
3. One-Page Exec Summary	Provide a clean business brief outlining recommendations and risks.	Focus on financial impact and lead time reductions.

8. Advanced Extensions

Students can expand the project's analytical depth with three technical extensions:

- **Geospatial Integration:** Incorporate geocoding to calculate exact Haversine distances between factory postal codes and customer destination coordinates, replacing simple regional categories with continuous spatial features.
- **Calibrated Cost Modeling:** Develop a dynamic logistics pricing matrix that estimates freight spend based on distance, volumetric weight, and carrier tiers.
- **Quantile Regression Modeling:** Replace single-point predictions with probabilistic models to forecast the 90th percentile of lead times, helping teams explicitly plan for worst-case supply chain delays.